Matthew Zhang

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Education

The University of Edinburgh | Edinburgh, UK *Expected May 2027*
Master of Informatics (MInf) in Computer Science
Current grade: First Class
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Security, Databases, Machine Learning, Natural Language Processing, Compiling Techniques

Experience

NASA | New York, US *June – August 2021*
Software Engineering Intern
- Built a Python pipeline to process and visualize earthquake precursor signals in 3D, enabling researchers to identify anomalies in seismic patterns more efficiently.
- Integrated the NASA WorldWind API into the analysis tool, giving scientists an interactive, globe-based interface for exploring geospatial data.
- Optimised simulation runtime by restructuring data pipelines, reducing average experiment processing time by 30%.

Edinburgh University Tutor | Edinburgh, UK *September 2023 – 2025*
- Tutored *Introduction to Computation* (Haskell) to first-year computer science students, clarifying core concepts in recursion, functional programming, and proof-based reasoning.
- Developed custom lesson plans tailored to each student's unique needs and learning styles.

Projects

High-Frequency Trading Simulator (Python, Pandas, NumPy)
- Implemented a limit order book simulator capable of processing thousands of orders per second.
- Implemented multiple trading strategies (mean reversion, momentum, arbitrage) and a backtesting framework to evaluate profitability under historical and synthetic market conditions.
- Added features for order book depth modelling and latency simulation.

Robotic Pool-Playing Interface (JavaScript, Node.js, Python)
- Developed a full-stack web application with an interactive drag-and-play virtual pool table that controls a physical robot in real time.
- Integrated computer vision for ball tracking and obstruction detection, with WebSocket communication for low-latency robotic control.
- Implemented a physics engine to handle collisions, shot trajectories, and automated turn-taking, enabling seamless human–robot gameplay.

Voice-Guided Video Segmentation for Microscopy (Python, PyTorch, OpenCV, Whisper, MicroSAM, SAM 2)
- Built a human-in-the-loop microscopy video segmentation system combining voice prompts and point-click interaction for expert-guided analysis.
- Integrated speech-to-text, intent parsing, frame segmentation, and temporal propagation into one end-to-end pipeline with correction and re-propagation workflows.
- Evaluated performance on 19 videos with 83 ground-truth frames across 9 experiments, including ablations, timing, and failure analysis.
- Improved temporal consistency substantially versus baseline (0.882 vs 0.114 mean score) and achieved 100% prompt-extraction accuracy on benchmark tests (10/10).
- Developed a reproducible experimentation and reporting pipeline producing CSV/JSON outputs plus publication-ready tables and figures.

Technical Skills

Languages: Python, Java, JavaScript, SQL, C, Haskell
Frameworks & Libraries: React.js, Node.js, Express, Pandas, NumPy, TensorFlow, PyTorch
Tools and Systems: Git/GitHub, Docker, Linux, ROS, REST APIs, WebSockets